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Calibration of Event Cameras Underwater

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Abstract

Event cameras represent a paradigm shift in robotic perception, offering microsecond temporal resolution and high dynamic range. However, their asynchronous nature renders traditional frame-based calibration algorithms ineffective. Deploying these sensors in maritime environments introduces further complexity, as the multi-media refraction through flat glass viewports invalidates the standard single-viewpoint camera model, inducing significant depth-dependent spatial distortion. Furthermore, while theoretical continuous-time calibration frameworks exist for event cameras, their monolithic batch-optimization architectures often exceed the memory constraints of standard field robotics hardware.

This thesis presents a highly optimized, continuous-time stereo calibration pipeline engineered specifically for underwater event cameras. To resolve memory bottlenecks, the architecture utilizes a sequential, sliding-window spatiotemporal tracking framework. To correct refractive distortion, the multi-media physics are explicitly embedded into a non-linear Ceres bundle adjustment using the Pinax model, stabilized by a novel structural coplanarity constraint. For real-time operational deployment, the optimized physical parameters are exported into dense Look-Up Tables (LUTs) paired with a sub-pixel bilinear splatting inversion module.

Empirical evaluation on unseen underwater datasets demonstrates that this proposed methodology significantly reduces the vertical epipolar disparity caused by asymmetric refractive shearing. Final validation via 3D point cloud metrology confirms that the pipeline accurately preserves metric scale and structural orthogonality. By neutralizing subsea optical distortion and capping computational memory growth, this work establishes a rigorous calibration foundation necessary for future high-frequency maritime visual odometry and metrology applications.

1 Introduction

The deployment of autonomous systems in subsea environments represents one of the most demanding operational frontiers in field robotics. Because marine environments are intrinsically GNSS-denied due to the rapid attenuation of electromagnetic radiation in water, Autonomous Underwater Vehicles (AUVs) rely entirely on dead reckoning frameworks to calculate state estimation and maintain localization during extended subsea missions [1]. Traditionally, these navigation suites utilize an Inertial Navigation System (INS) paired with an acoustic Doppler Velocity Log (DVL) [1]. The DVL functions by emitting acoustic beams toward the seafloor and computing the vehicle’s metric linear velocity relative to the bottom via the Doppler frequency shift effect. While highly precise in open, unobstructed water columns over flat seabeds, these acoustic sensors suffer significant performance degradation in complex, close-range structural environments [2]. Within enclosed spaces, near vertical reef structures, or during overflights of highly reflective man-made infrastructure, DVLs are prone to severe multipath echoes, signal absorption from soft marine sediments, and an intermittent loss of bottom lock. Consequently, navigation accuracy degrades precisely within the high-risk zones where precision control is most critical, such as autonomous docking or close-range shipwreck inspections [3].

Underwater visual odometry using frame-based cameras offers a high-resolution, passive alternative for close-range perception. However, traditional vision sensors struggle with rapid motion blur from turbulent marine currents, low spatial contrast, and severe caustic flickering near the surface. Event-based vision, driven by Dynamic Vision Sensors

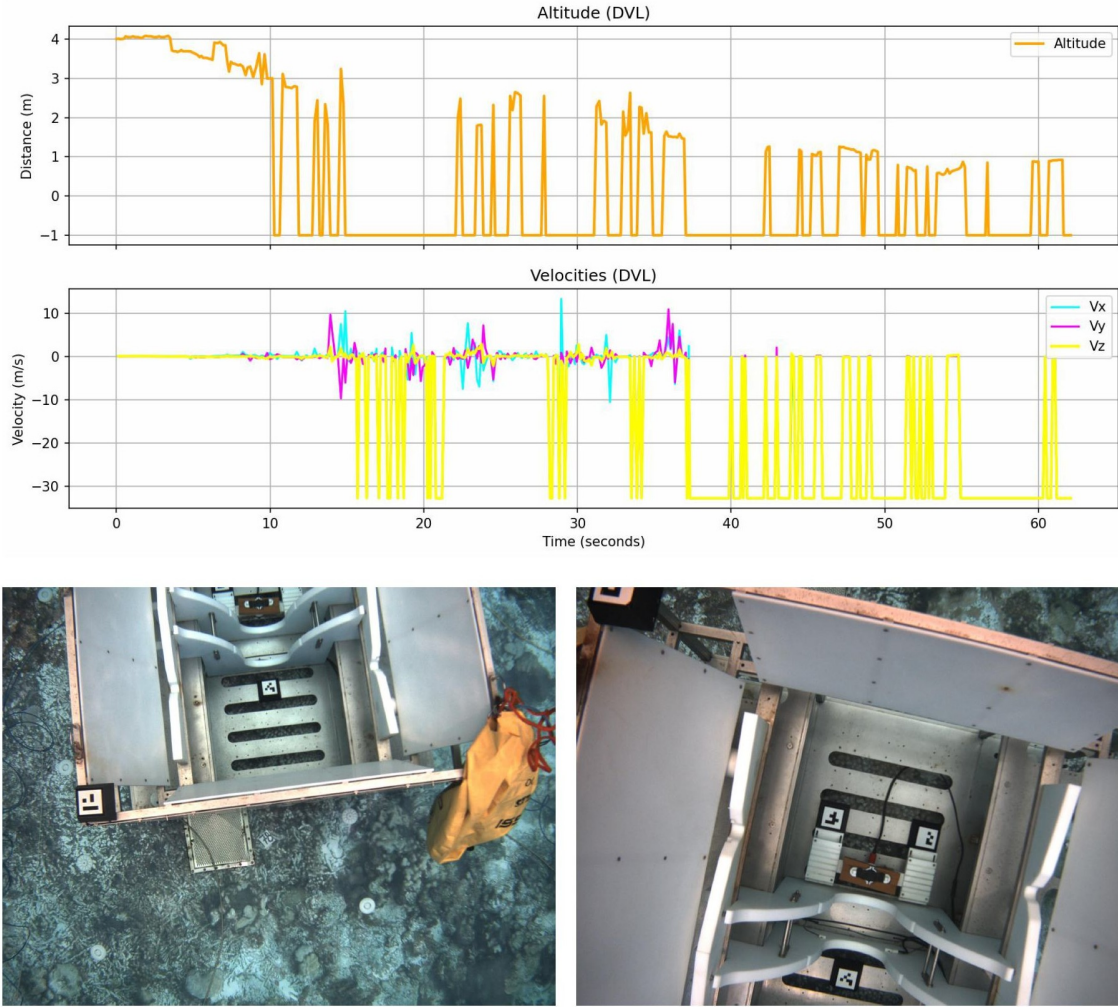


Figure 1: Illustration of DVL failure during overflight of a docking station: Time series of DVL-reported altitude and DVL-reported velocities (top); camera image at approximately $t=6s$, marking the onset of the overflight, when the docking structure first enters the DVL sensing region and the measurement quality begins to deteriorate (bottom-left); camera image at approximately $t=18s$, when the vehicle is further over the docking station (bottom-right).

(DVS), offers a transformative paradigm shift by discarding fixed-frame integration in favor of independent, pixel-level asynchronous brightness sampling [4]. This operational mechanism yields microsecond-scale temporal resolution, a high dynamic range exceeding 120 dB capable of surviving simultaneous deep-sea darkness and surface glare, and the complete elimination of motion blur.

To harness event-based vision for absolute metric tasks such as optical velocity estimation or dense 3D mapping, the sensors must be deployed in a stereoscopic configuration to recover scale without monocular ambiguity [5]. However, utilizing stereo event cameras in maritime environments introduces a fundamental prerequisite: the sensor pair must be accurately and rigorously calibrated to resolve both their internal optics and their relative spatial geometry.

1.1 Problem Statement

Unlocking the analytical potential of stereo event cameras underwater requires overcoming a rigid “double barrier” of sensor calibration. First, standard terrestrial calibration algorithms designed for synchronous frame snapshots cannot ingest an asynchronous spatiotemporal stream of events. While commercial baselines, such as frequency-filtering pipelines, circumvent this by accumulating frames from active blinking targets [6], they inherently sacrifice the continuous-time precision of the sensor and remain highly susceptible to motion blur during active maneuvers.

Second, placing the camera rig inside a watertight pressure housing introduces severe refractive distortions governed by Snell’s Law [7]. As light rays travel from the water medium into a flat-pane glass viewport and cross into the internal air enclosure of the housing, the single-viewpoint property of the vision sensor is fundamentally broken [1, 2]. Uncorrected pinhole camera models operating in multi-media environments systematically miscalculate depth fields, yielding scaling errors where subsea structures are estimated to be approximately 33% closer and larger than their true physical dimensions [7].

While continuous-time parameter estimation methods have successfully modeled event streams in air using parametric B-splines, mapping these frameworks directly to underwater environments presents a significant computational bottleneck. State-of-the-art continuous-time calibration solvers process spatial trajectories efficiently in theory, but their global batch-optimization architectures scale superlinearly in memory consumption. This structural overhead frequently causes standard field-deployment hardware to exhaust available memory when handling dense operational datasets.

Consequently, there is a critical need for a localized, memory-efficient reimplementation that explicitly embeds multi-media refraction physics into a continuous-time tracking framework. Furthermore, validating the metric truth of such an underwater calibration pipeline requires more than relative pixel residuals. This thesis addresses these challenges by engineering an optimized refractive calibration pipeline and verifying its accuracy through absolute 3D metrology. By accumulating asynchronous event packets back-projected through dense refractive Look-Up Tables (LUTs), the framework isolates subsea laser light structures. The resulting unwarped structural profiles are triangulated into metric 3D point clouds. By scanning objects of known geometry and running half-space bounding volume intersections to segment planar features, the structural reconstruction is directly evaluated for orthogonality and absolute edge lengths, providing strong empirical evidence of the calibration pipeline’s ability to preserve rigid spatial consistency subsea.

2 Background

Standard frame-based cameras capture dense, synchronous snapshots of the visual field at a fixed, global frame rate (e.g., 30 or 60 Hz). This global exposure mechanism creates fundamental operational limitations for field robotics: in highly dynamic environments, fast-moving objects induce severe motion blur, and the restricted dynamic range (typically 60 dB) frequently results in overexposure or underexposure when navigating environments with extreme lighting gradients. In stark contrast, event cameras—such as Dynamic Vision Sensors (DVS)—operate on a fundamentally different, biomimetic paradigm inspired by the human retina [8]. Rather than capturing fixed frames, each

pixel in an event camera operates independently and asynchronously, reporting a localized spike (an "event") only when it detects a change in log-level brightness that exceeds a predefined threshold.

This asynchronous, sparse data generation yields three critical advantages: microsecond-level temporal resolution that virtually eliminates motion blur, an exceptionally High Dynamic Range (>120 dB) capable of simultaneously resolving dark shadows and bright glare, and highly efficient data throughput since redundant, no need to track pixel information that is unchanged [8].

Consequently, stereo event cameras have recently revolutionized autonomous navigation and 3D perception in highly dynamic or visually degraded terrestrial environments. For example, Zhou et al. [9] developed the Event-based Stereo Visual Odometry (ESVO) pipeline, which utilizes stereo event cameras to perform real-time, semi-dense 3D mapping and 6-DOF tracking by extracting temporal consistency directly from raw event streams. To push the boundaries of autonomous driving in challenging lighting conditions (such as nighttime driving or direct sun glare), Gehrig et al. [10] introduced DSEC, a large-scale automotive dataset captured with high-resolution stereo event sensors. Similarly, Zhu et al. [11] presented the MVSEC dataset, demonstrating the viability of stereo event configurations mounted on highly agile robotic platforms, including hexacopters and fast-moving handheld rigs, for high-speed Simultaneous Localization and Mapping (SLAM).

Despite these rapid advancements in terrestrial robotics, the deployment of stereo event cameras in subsea environments remains an entirely unexplored frontier. The physical characteristics of the underwater medium—such as rapid motion blur from turbulent currents, severe caustic flickering, and extreme low-light constraints—perfectly align with the inherent hardware strengths of dynamic vision sensors. However, they have not yet been successfully utilized underwater because doing so fundamentally requires precise multimedia sensor calibration. Modeling the complex flat-plane refractive optics of a watertight pressure housing specifically for asynchronous spatiotemporal event streams is a massive mathematical hurdle that must be overcome before any subsea event-based tracking or odometry can be achieved. Thus, engineering a framework capable of calibrating these sensors underwater is the primary novel proposition of this thesis, establishing a foundational requirement for this technology's future in marine robotics.

3 Related Work

The endeavor to accurately calibrate event cameras, particularly in challenging marine environments, sits at the intersection of two historically distinct research domains: asynchronous sensor calibration and multi-media optical refraction. Because event cameras discard the concept of fixed-frame image capture in favor of continuous, microsecond-level brightness tracking [4], legacy calibration toolboxes designed for standard pinhole cameras are rendered mathematically obsolete. This chapter reviews the chronological evolution of event-based calibration—from early frame-reconstruction paradigms to modern first-principles physics solvers—and surveys the state-of-the-art in flat-plane refractive modeling required for underwater deployment.

3.1 The Reconstruction Paradigm

Initial attempts to bridge the gap between asynchronous event streams and traditional calibration algorithms focused on forcing events back into the spatial domain of discrete frames. Muglikar et al. [3] introduced E2Calib, a framework that utilizes a neural network (E2VID) to reconstruct video frames from accumulated event slices. These reconstructed images are subsequently processed by standard corner detectors to identify passive checkerboard patterns. While E2Calib eliminated the need for active lighting, the neural reconstruction introduced quantization artifacts and temporal smearing that degraded sub-pixel precision [3].

Conversely, the industry-standard Metavision Intelligence Suite [6] relies on a frequency-filtering approach. By utilizing an active LCD screen displaying a blinking checkerboard pattern, the Metavision pipeline isolates events triggered strictly at the target’s blinking frequency. While computationally lightweight, this method mandates that the camera remains perfectly static; rapid camera motion induces temporal blur across the accumulated frames. Furthermore, carrying active, waterproof LCD screens into field environments for underwater calibration is operationally restrictive.

3.2 Direct Geometric and Physics-Based Calibration

To bypass the interpolation artifacts inherent in image reconstruction, recent frameworks process the raw event stream directly. Traditional standard pinhole camera calibration relies on the Brown-Conrady distortion model, which maps an idealized normalized image coordinate (x, y) to a distorted pixel coordinate (x_d, y_d) via radial and tangential coefficients:

$$x_d = x(1 + k_1r^2 + k_2r^4 + k_3r^6) + 2p_1xy + p_2(r^2 + 2x^2) \quad (1)$$

where $r^2 = x^2 + y^2$. Frameworks such as E-Calib [12] attempt to solve for these coefficients by processing raw event streams triggered by a static circle grid, employing a Reweighted Least Squares algorithm to fit ellipses to event clusters. However, grouping events into discrete time windows inherently caps the temporal resolution of the sensor.

The current state-of-the-art in terrestrial event calibration discards time-windows entirely in favor of continuous-time physics. Chen et al. introduced eKalibr [4], a framework that models the physical generation of events from first principles. By assuming that events generated by moving edges form local planes in continuous $x - y - t$ space, the local spatiotemporal plane is defined as:

$$Ax + By + Ct + D = 0 \quad (2)$$

From this planar geometry, eKalibr extracts the normal optical flow $\mathbf{v}_n$ directly from the event stream without relying on spatial gradients:

$$\mathbf{v}_n = -\frac{1}{C} \begin{bmatrix} A \\ B \end{bmatrix} \quad (3)$$

By using this velocity vector, eKalibr dynamically warps events backward in time to mathematically “freeze” motion blur before fitting spatial parameters [4].

This formulation was subsequently expanded in eKalibr-Stereo [5], which models the camera trajectory as a Continuous-Time B-Spline. By jointly optimizing spatial reprojection error and temporal hardware offsets, eKalibr-Stereo achieves sub-millimeter baseline accuracy and microsecond temporal alignment without relying on physical synchronization cables.

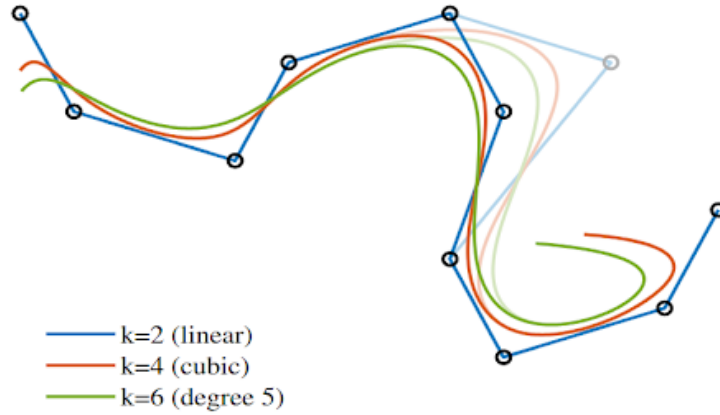


Figure 2: Illustration of continuous-time B-Spline trajectory modeling parameterized by discrete control knots (black circles). The colored curves represent continuous spatial trajectories of varying polynomial degrees ($k = 2, 4, 6$). In continuous-time calibration frameworks, this mathematical representation allows the solver to query the exact microsecond-level pose of the camera along the smooth trajectory between discrete control points. (Adapted from Sommer et al. [13]).

3.3 Multi-Media Refractive Optics

Deploying optical sensors subsea requires confining the hardware within watertight pressure housings equipped with flat glass viewports. The standard pinhole projection equations break down in this environment because the multi-layer flat refractive geometry of the air-glass-water interfaces causes light rays to bend according to Snell’s Law [1]:

$$n_{\text{water}} \sin(\theta_1) = n_{\text{glass}} \sin(\theta_2) = n_{\text{air}} \sin(\theta_3) \quad (4)$$

This multi-media bending invalidates the single-viewpoint property of the camera, inducing non-linear distortion and severe depth magnification [1].

Jordt et al. [2] addressed this by formulating a full Refractive Camera Model that explicitly traces individual light rays through the glass interface. Their non-linear optimization pipeline solves for both the camera’s internal geometry and the physical orientation of the housing pane (surface normal tilt and lens-to-glass distance). While Jordt’s framework achieves metrology-grade accuracy, solving 12th-degree polynomial roots for every ray is computationally prohibitive for real-time robotic deployment [2].

To bridge the gap between rigorous physics and real-time computation, Łuczyński et al. [14] proposed the Pinax (Pinhole + Axial) model. Recognizing that refractive distortion through flat ports exhibits radial symmetry, the Pinax model approximates the multi-media ray bending as a dynamic shift of the camera’s optical center along the Z-axis [14]. This allows the complex refractive geometry to be pre-computed into highly efficient Look-Up Tables (LUTs). More critically for field robotics, the Pinax formulation enables “dry-to-wet” calibration transfer—where precise in-air camera intrinsics can be mathematically warped to predict underwater distortion without requiring a cumbersome subsea checkerboard calibration session [15].

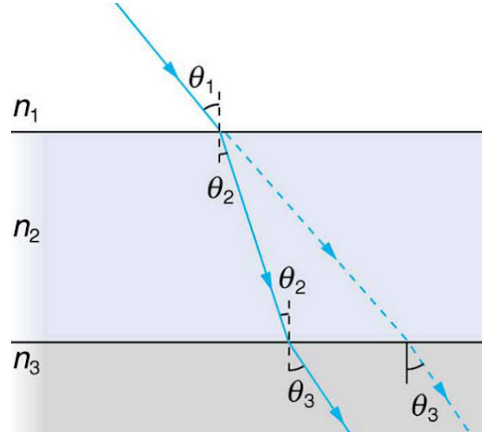


Figure 3: Multi-media optical raytracing diagram. Light rays originating in water ($n_1 \approx 1.33$) bend at the glass port ($n_2 \approx 1.49$) before entering the air enclosure ($n_3 \approx 1.0$), invalidating standard single-viewpoint pinhole geometry.

3.4 Summary

While continuous-time parameter estimation (eKalibr) offers the mathematical rigor required to accurately calibrate asynchronous stereo events, and models like Jordt’s and Pinax address the physics of multi-media refraction, no existing framework unifies these domains efficiently for online subsea deployment. The underwater calibration of event-based vision sensors is an entirely novel domain; to date, there are no existing methodologies or literature addressing this specific challenge. The subsequent chapters detail the computational bottlenecks inherent in these foundational terrestrial works and establish the engineering necessity for a localized, highly optimized reimplementation capable of bridging this gap for the first time.

3.5 Thesis Contributions and Structural Outline

This thesis systematically addresses the dual challenge of asynchronous data processing and multi-media refractive correction for event sensors. The primary contributions of this work are the implementation, expansion, and computational optimization of an underwater calibration pipeline utilizing state-of-the-art continuous-time mathematics alongside the explicit modeling of flat-plane refraction geometry. Relying on the multi-layer flat refractive foundations established by Agrawal et al. [1], our architecture transforms non-single-viewpoint optimization constraints into structured, parallelized tracking updates.

4 Engineering Motivations for Reimplementation

While state-of-the-art continuous-time frameworks like eKalibr [4] and eKalibr-Stereo [5] provide a mathematically rigorous foundation for event-camera calibration in air, attempting to deploy these frameworks for operational subsea robotics reveals severe computational bottlenecks. This chapter analyzes the performance and memory constraints of existing batch-optimization paradigms and details why a localized, lightweight reimplementation was necessary for practical maritime field deployment.

4.1 Constraints of Maritime Field Deployment

Calibrating an Autonomous Underwater Vehicle (AUV) visual pipeline is rarely performed in a controlled, pristine laboratory environment. In practice, recalibration must often be executed onboard a research vessel or dockside just prior to deployment. Consequently, the computational hardware available is typically restricted to ruggedized field laptops or mobile workstations, which are constrained by battery power, thermal throttling, and limited Random Access Memory (RAM)—typically capped between 16 GB and 32 GB.

Deploying complex mathematical frameworks under these conditions requires software that is highly optimized for memory efficiency. If an algorithm requires high-performance desktop-grade hardware (e.g., 64 GB to 128 GB of RAM) to process a standard 30-second calibration sequence, it becomes fundamentally useless for field robotics. As demonstrated in the subsequent profiling tests, the original eKalibr architecture rapidly exceeded standard field hardware capabilities, resulting in terminal memory exhaustion (segmentation faults) before optimization could even begin.

4.2 Architectural Causes of Memory Exhaustion

An analysis of the foundational continuous-time solvers reveals three primary architectural decisions that drive exponential memory consumption: monolithic bag processing, dense spatiotemporal grid tracking, and unbounded B-Spline control point generation.

4.2.1 ROS 1 Bag Parsing vs. ROS 2 Sequential Streaming

The original eKalibr framework is deeply coupled with the Robot Operating System 1 (ROS 1) ecosystem. ROS 1 bag parsing architectures traditionally index and load vast swaths of message data into memory trees prior to computation. Because event cameras generate data asynchronously at extremely high throughputs (often exceeding millions of events per second), buffering these raw `EventPacket` messages in a ROS 1 environment forces the system to cache gigabytes of data overhead simply to read the file.

To resolve this, our reimplementation transitions the entire pipeline to a ROS 2 architecture, explicitly utilizing a sequential bag reader within ROS 2. This allows the system to stream raw event packets sequentially, decoding and processing them in isolated sliding windows rather than loading the monolithic dataset into RAM.

4.2.2 Spatiotemporal Grid Tracking

To extract the normal optical flow of moving edges, standard first-principles solvers utilize a dense spatiotemporal voxel grid to group events. In eKalibr, generating the constraints for the Ceres Solver [?] requires maintaining this grid across the entire temporal length of the calibration sequence. As the temporal length of the sequence increases, the memory footprint of this grid grows linearly, rapidly consuming available heap space. Our reimplementation abandons this global grid approach in favor of localized, asynchronous event queues and lightweight temporal decay functions (e.g., the Time Surface processing module) that gracefully drop historical data once it falls outside the active optimization window.

4.2.3 Continuous-Time B-Spline Creation

The core advantage of continuous-time calibration is the parameterization of the camera trajectory using cumulative B-Splines on the Lie group $SE(3)$ manifold. As demonstrated by Sommer et al. [13], formulating the trajectory via cumulative B-Splines ensures smooth, continuous pose interpolation while enabling highly efficient analytical derivative computation during non-linear optimization. This mathematical formulation allows the solver to query the exact camera pose at the precise microsecond an asynchronous event is generated. However, to capture the high-frequency micro-vibrations of a moving camera, the B-Spline requires a high density of temporal “knots” (control points).

In a batch-optimization framework, every single control point generated across the entire calibration sequence is pushed into a massive Jacobian matrix for the Ceres bundle adjustment. For a 60-second calibration video recorded at high temporal resolution, this generates thousands of $SE(3)$ control variables. The memory required to allocate and invert the resulting Hessian matrix during the Levenberg-Marquardt optimization phase causes immediate memory swapping and eventual system crashes on standard field laptops.

4.3 Memory Profiling and Performance Results

To quantify these limitations, memory profiling was conducted comparing the baseline eKalibr architecture against our sequentially optimized pipeline. Calibration sequences ranging from 0 to 60 seconds in 10-second intervals were processed, and peak RAM allocation was recorded.

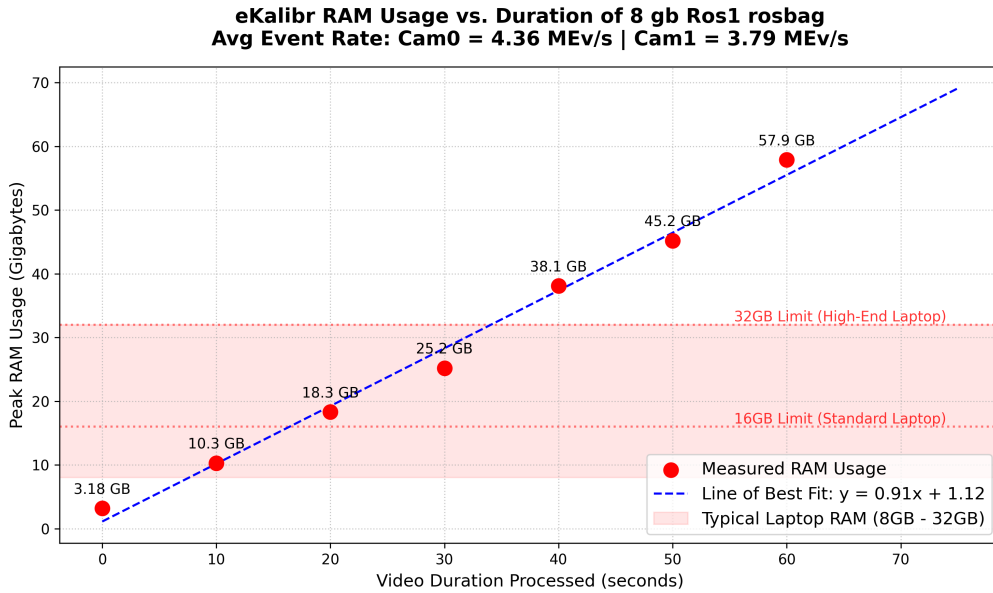


Figure 4: Peak RAM consumption relative to the duration of the calibration sequence. The legacy eKalibr (ROS 1 / B-Spline) architecture scales exponentially, crossing the 32 GB threshold at approximately 30 seconds. The proposed ROS 2 sequential pipeline maintains an under 16 GB memory footprint for 60 seconds.

As illustrated in Figure 4, the baseline continuous-time framework exhibits a linear-to-exponential memory growth, frequently requiring upwards of 60 GB of RAM for a standard 60-second sequence. This is a direct consequence of the global grid tracking

and the massive B-Spline control point matrix. By decoupling the ROS 1 message overhead and relying on sliding-window sequential processing, our reimplementation caps the memory footprint, ensuring the algorithm remains viable on a standard 16 GB marine field laptop for a 60 second data capture.

4.4 Summary

The transition from a theoretical framework to a field-ready operational pipeline necessitated stripping away global batch-optimization paradigms. By shedding the ROS 1 legacy overhead, avoiding full-sequence grid tracking, and capping B-Spline knot generation via sequential processing windows, we engineered a system capable of executing state-of-the-art continuous-time mathematics within the rigid hardware constraints of maritime robotics. The following chapter details the exact software architecture and mathematical implementations used to achieve this localized processing pipeline.

5 Calibration Setup and Hardware Architecture

To empirically validate the proposed continuous-time calibration and multi-media refractive optimization frameworks, a custom stereoscopic event camera rig was engineered. This section details the physical hardware components, the data acquisition architecture, and the operational trade-offs encountered regarding sensor biasing and memory bandwidth during subsea deployment.

5.1 Stereo Vision Hardware and Physical Rig

The primary perception payload consists of a synchronized stereoscopic pair of Prophesee EVK4HD event cameras. These sensors were rigidly mounted to a custom structural baseline to ensure strict mechanical stability during highly dynamic maneuvers 5. Because continuous-time calibration relies on microsecond-level temporal alignment, the two cameras were physically genlocked using a hardware synchronization cable, designating one sensor as the Master and the other as the Slave, ensuring their internal clocks remained perfectly coupled during operation.

5.2 Data Processing and Acquisition Architecture

Given the immense data throughput of asynchronous event sensors, routing raw event streams directly to a standard computing unit is often inefficient and prone to USB bus saturation. To manage this, the hardware architecture integrates an AMD Kria KV260 Vision AI Starter Kit as an intermediary edge-processing module.

The Kria board directly interfaces with the stereo cameras via USB 3.0, handling the initial spatiotemporal preprocessing and hardware driver execution. However, because the Kria board possesses a limited 4 GB of Random Access Memory (RAM), it cannot locally store the massive datasets required for a full continuous-time calibration sequence. To circumvent this, the Kria is networked via a high-speed Gigabit Ethernet connection directly to a host field laptop. As the Kria ingests the raw asynchronous event packets, it immediately streams the data over the local network to the laptop, which records the stream directly to disk as a unified ROS 2 bag (MCAP format).

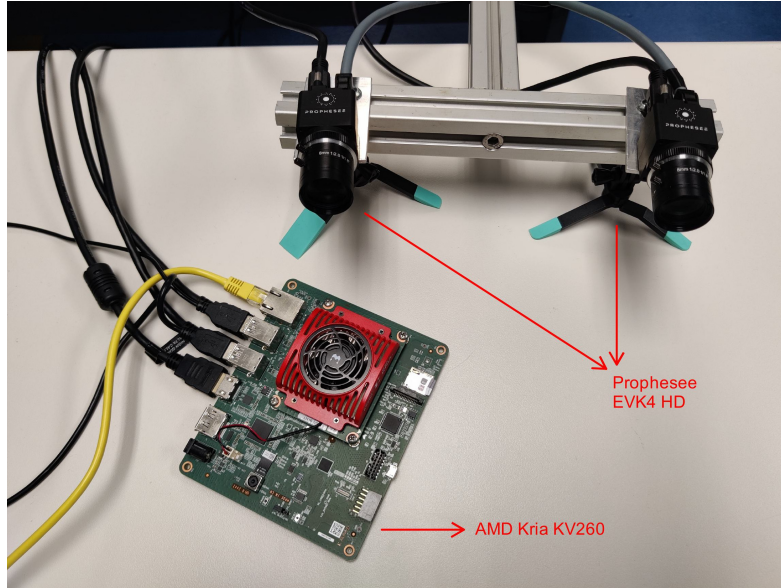


Figure 5: Annotated hardware diagram of the AMD Kria KV260 processing unit, highlighting the USB 3.0 sensor inputs and the high-speed Ethernet data pipeline routing to the host laptop.

5.3 Sensor Biasing and Memory Bandwidth Constraints

During the evaluation of the hardware pipeline, a critical operational bottleneck was identified regarding the sensors’ internal biasing configuration. Event cameras operate by triggering events only when a log-level change in brightness exceeds a specific threshold (the bias). By lowering these threshold biases, the sensors become significantly more sensitive, capturing micro-variations in contrast and extremely fast motion with pristine clarity.

However, this increased sensitivity inherently triggers an exponential surge in the overall event rate ⁶. In visually noisy marine environments—characterized by moving particulates, backscatter, and caustic lighting—a lowered bias setting can cause the sensors to generate tens of millions of events per second. During hardware evaluation, it was observed that if the resulting data spike exceeds the transmission bandwidth of the Gigabit Ethernet connection, the un-transmitted event packets begin to queue within the Kria board’s local memory. Under high-sensitivity bias configurations, this rapid queuing can entirely exhaust the Kria’s 4 GB of RAM within seconds, leading to immediate system failure and dropped data packets. Consequently, the sensor biases must be carefully tuned to balance optimal geometric sensitivity against the strict throughput limitations of the edge hardware.

5.4 Calibration Target

To provide the rigorous geometric ground truth required by the continuous-time spatiotemporal solver, a custom asymmetric circle grid ⁷ was utilized as the primary calibration target. Unlike standard checkerboards, which can suffer from the aperture problem during linear motion detection, the circular features provide uniform spatial gradients in all directions. The asymmetry of the grid ensures that the continuous-time solver can unambiguously resolve the 6-DOF orientation of the target plane at any given microsecond, regardless of the board’s rotation within the multi-media environment.

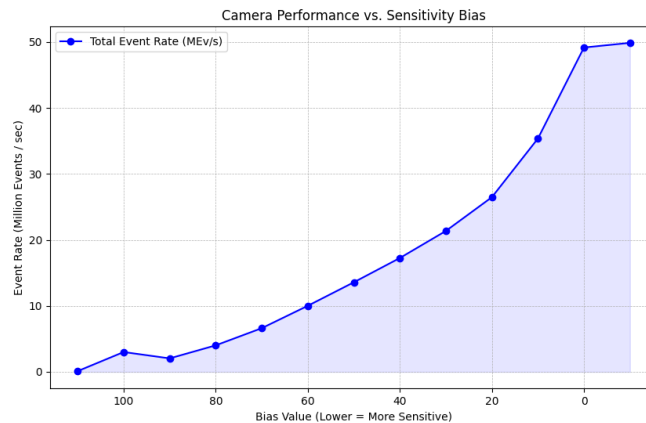


Figure 6: Hardware evaluation plot illustrating the relationship between sensor bias thresholds, resulting higher event rates. Lowering the biases maximizes sensitivity but rapidly fills the Kria board’s 4 GB RAM buffer if the Ethernet offload bandwidth is exceeded.

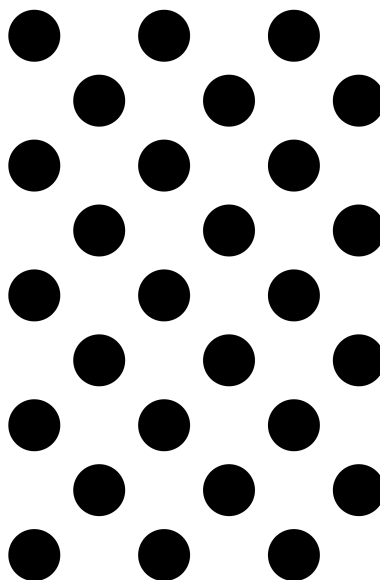


Figure 7: The asymmetric circle grid target utilized for the continuous-time spatiotemporal calibration. The asymmetric layout guarantees unambiguous 3D pose estimation throughout the recording sequence.

6 Methodology

As established in Section 4, the transition from theoretical event calibration to a field-ready operational architecture necessitates restructuring the computational paradigm. Rather than relying entirely on monolithic memory caching, our methodology decouples the data pipeline: initial feature extraction is executed within a localized, sequential sliding window, which then feeds an aggressively filtered dataset into a final geometric global batch-optimization. This section details the architectural implementations of our monocular calibration pipeline, bridging asynchronous data tracking, statistical noise rejection, and multi-media refractive optics.

6.1 Monocular Event Ingestion and Temporal Alignment

The foundation of the pipeline fundamentally restructures how asynchronous data is handled in memory. Bypassing the heavy message-caching overhead of legacy ROS 1 tools, the system employs ROS 2’s sequential bag reader threaded architecture to stream raw `EventPacket` messages offline.

Because event sensors stream data asynchronously at microsecond resolution, hardware clock rollovers and packet desynchronization are common, especially in high-activity underwater scenes. The ingestion thread utilizes a strict temporal healer that decodes individual event polarities and aligns their raw hardware timestamps (t) to a unified, monotonically increasing system time. This guarantees that as the data is buffered into localized microsecond queues (e.g., sliding windows of 20 to 50 ms), the temporal topology remains unbroken prior to geometric extraction.

6.2 Spatiotemporal Planar Fitting and Edge Classification

To extract geometric structure from this continuous stream, the pipeline estimates the localized trajectory of moving edges. When an active target, such as an asymmetric circle grid, moves across the sensor’s field of view, the events generated by a single geometric edge form a local plane in the spatiotemporal domain. For a localized cluster of events, the algorithm fits a continuous-time plane defined by the normal vector $\mathbf{n} = [n_x, n_y, n_t]^T$:

$$n_x x + n_y y + n_t t + d = 0 \quad (5)$$

From this planar equation, the normal optical flow is derived using a RANSAC-based estimator. The normal velocity vector $\mathbf{v}_n = [v_x, v_y]^T$ is extracted as:

$$\mathbf{v}_n = -\frac{n_t}{n_x^2 + n_y^2} \begin{bmatrix} n_x \\ n_y \end{bmatrix} \quad (6)$$

Crucially, rather than warping these events into an intermediate 2D motion-compensated frame—which can reintroduce spatial quantization errors—our pipeline utilizes this normal flow purely as a directional heuristic. By analyzing the velocity vectors, the system accurately classifies the spatiotemporal clusters into leading and trailing edges. This classification securely pairs opposite-polarity clusters belonging to the same physical calibration circle, allowing the pipeline to pass their raw (x, y, t) coordinates directly into the 3D continuous-time ellipse solver without relying on flattened image representations.

6.3 Statistical Outlier Rejection for Noise

Marine environments are visually chaotic. During subsea calibration, the event sensor inevitably triggers on severe background noise, including moving marine snow, dense particulate matter, and large caustic specular flashes off the flat glass viewport. Attempting to fit geometries to these chaotic event clusters rapidly causes solver divergence.

To mitigate this, our implementation introduces a dynamic, statistics-based median-size noise filter. Once the initial spatiotemporal clusters are formed, the pipeline calculates the median topological size of the detected groupings. Using the known dot count of the calibration target as a heuristic baseline, the system surgically slices off extreme outliers. Massive event groupings generated by lightning-fast light reflections (which form artificially large clusters) and infinitesimal noise specks are aggressively pruned from the active memory pool. This ensures that only uniformly sized clusters physically representing the calibration grid are passed to the downstream optimization.

6.4 Interactive FOV Profiling and Grid Review

Transitioning from a fully automated calibration to human-in-the-loop verification is a core requirement for field robotics. Optimizing over a dataset that is heavily skewed to one side of the image plane inevitably results in severe radial distortion overfitting on the unconstrained periphery of the lens.

Before committing any detected geometries to the global solver, the proposed pipeline suspends execution and launches an **Interactive Grid Review Phase**. The operator is presented with a sequential visual playback of the extracted target geometries. The interface overlays a bounding convex hull around the current grid and dynamically updates a cumulative 2D **Field of View (FOV) Coverage Map**.

This coverage map acts as a spatial histogram. As grids are manually approved by the operator, the corresponding spatial bins increment on the display. This allows the user to actively shape the spatial distribution of the calibration data, ensuring dense, uniform coverage across the entire lens. If an operator identifies a grid that passed algorithmic filtering but is visually corrupted by a falsely classified ellipse, they can manually reject it. This review phase guarantees that the final optimization dataset is spatially diverse and free from localized refractive anomalies.

6.5 Stereo Event Ingestion and Temporal Healing

Transitioning from monocular calibration to a binocular framework introduces the critical challenge of temporal synchronization. In standard frame-based stereo, cameras share a hardware trigger to snap images simultaneously. Event cameras, however, stream independent, asynchronous data packets. If the left and right event streams are misaligned, a rapidly moving calibration target will occupy physically different spatial locations in each sensor’s timeline, catastrophically corrupting the stereo baseline estimation.

To resolve this, the pipeline utilizes a rigorous offline buffer tightly coupled with a custom event processor temporal healer. Rather than dynamically searching for matching timestamps between the two streams, the temporal healer actively decodes raw sensor timestamps and corrects hardware clock rollovers on the fly. This safely locks the asynchronous events of both cameras to a unified, monotonically increasing ROS wall-clock time.

Because both event streams are rigidly locked to this global clock, the pipeline safely and synchronously slices the data into fixed absolute-time windows (e.g., 20 ms intervals). This eliminates the need for dynamic tolerance searching and guarantees that any spatial geometry extracted from the left camera has an exact, microsecond-aligned counterpart in the right camera’s memory buffer.

6.6 Downstream Post-Calibration Rectification: Bilinear Splatting

With the static camera geometries successfully solved by the calibration pipeline, the system transitions to a dedicated downstream evaluation node utilizing the event rectifier module to validate the mathematical integrity of the calibration.

Traditional OpenCV rectification relies on backward mapping (pulling pixels from a distorted source to a clean destination). However, applying standard pull-mapping to sparse, continuous event coordinates induces severe quantization artifacts. Because event sensors produce mathematically sparse point clouds rather than dense pixel arrays, backward mapping forces fractional event coordinates into discrete pixel bins, resulting in jagged spatial lines that severely degrade metrology evaluation.

To preserve the sub-pixel continuity of the event stream, our event rectifier module implements a custom **Bilinear Splatting Push-Map Inverter**. Instead of querying where a destination pixel came from, the algorithm “pushes” the raw event coordinate (u, v) forward through the distortion field to its rectified sub-pixel location (x, y) . The intensity (or weight) of the event is then distributed bilinearly across the four nearest integer pixel coordinates:

$$W_{\lfloor x \rfloor, \lfloor y \rfloor} = (1 - \Delta x)(1 - \Delta y) \quad (7)$$

$$W_{\lceil x \rceil, \lfloor y \rfloor} = \Delta x(1 - \Delta y) \quad (8)$$

$$W_{\lfloor x \rfloor, \lceil y \rceil} = (1 - \Delta x)\Delta y \quad (9)$$

$$W_{\lceil x \rceil, \lceil y \rceil} = \Delta x\Delta y \quad (10)$$

where $\Delta x = x - \lfloor x \rfloor$ and $\Delta y = y - \lfloor y \rfloor$. This forward-splatting technique guarantees a perfectly smooth, sub-pixel gradient across the rectified image plane, effectively eliminating geometric aliasing before stereo matching occurs. (*Note: While the core calibration pipeline generates dense multi-media LUTs, the integration of these specific underwater refractive maps into the real-time splatting module remains a target for future operational deployment, as detailed in Chapter 7*).

6.7 Spatiotemporal Grid Extraction and the Four-Stage Geometric Filter

Once the continuous-time ellipse fitting has isolated the sub-pixel centers of the calibration target, the system must topologically reconstruct the asymmetric circle grid. However, relying solely on basic blob detection in a multi-media marine environment is highly susceptible to false positives generated by caustic flashes or refractive warping. To guarantee that only mathematically pristine geometries are passed to the bundle adjustment, our methodology implements a strict sequential verification process composed of four distinct geometric gates.

For a spatiotemporal event cluster to be validated as a true calibration grid, it must successfully pass through all four gates:

1. **Gate 1 (Visibility and Blob Count):** The initial event canvas is evaluated by a spatial blob detector to ensure the exact number of required calibration circles (e.g.,

27 discrete dots) are clearly visible and spatially distinct. If the detector registers a different count due to occlusion or lighting noise, the window is instantly discarded.

2. **Gate 2 (Topological Geometry):** The surviving centers’ structural topology is verified, ensuring the points conform to the strict spacing, angles, and asymmetric layout of the physical calibration board.
3. **Gate 3 (3D Pose Estimation):** For topologically valid grids, the pipeline employs a Perspective-n-Point algorithm. Using an initial guess of the camera intrinsics, the solver attempts to mathematically map the 2D pixel detections to an idealized 3D mathematical model of the board. If the solver fails to converge on a physically possible pose, the grid is rejected.
4. **Gate 4 (Reprojection Threshold):** The final and most restrictive gate calculates the localized Root Mean Square (RMS) reprojection error of the 3D pose estimated in Gate 3. If the RMS exceeds a strict threshold (typically set to 2.0 or 3.0 pixels), the grid is classified as a refractive outlier and is permanently purged.

Only grids that successfully navigate this rigid four-stage filtering process are queued for the Interactive Review Phase and subsequent non-linear stereo optimization.

6.8 Multi-Stage Refractive Bundle Adjustment

Standard in-air stereo calibration frameworks utilize non-linear optimization to minimize the spatial reprojection error between the mathematical Brown-Conrady distortion model [16, 17] and the observed 2D detections. However, applying this standard single-viewpoint mathematics to a camera enclosed within a flat-pane glass housing results in severe systematic drift [1]. To accurately model the multi-media bending of light rays through the air-glass-water interfaces, our methodology explicitly embeds flat-pane refractive physics directly into the bundle adjustment.

To prevent the optimizer from diverging due to the highly coupled nature of refractive depth scaling and stereo translation, the optimization is decoupled into a strict two-stage process: viewport initialization and joint stereo extrinsic optimization. Crucially, throughout both phases, the pre-calibrated in-air camera intrinsics and distortion coefficients are strictly locked. This forces the optimizer to channel all geometric corrections exclusively into the stereo baseline and the refractive multi-media parameters, drastically increasing the mathematical stability of the solver.

6.8.1 Phase 1: Agrawal Viewport Estimation and the Coplanarity Constraint

The primary source of non-linear distortion in subsea enclosures is the physical orientation of the glass port relative to the camera’s optical center. Relying on the multi-layer flat refractive foundations established by Agrawal et al. [1], the first phase of the solver is dedicated exclusively to isolating the physical “air gap” and the surface normal tilt of the camera housing.

The camera’s intrinsic matrix $\mathbf{K}$ and distortion coefficients $\mathbf{D}$ are locked to their pre-calibrated in-air values. To prevent the optimization of physically impossible geometries, the parameter space of the glass plane is strictly bounded. By mathematically locking

the Y and Z components of the surface normal, the solver is restricted to optimizing only the pitch angle and the physical orthogonal distance (d) from the lens to the glass.

Crucially, this phase drastically reduces the optimization dimensionality by eliminating the need to independently solve for the right camera’s viewport parameters. Because both stereo cameras are housed behind a single, continuous flat glass port, treating the left and right refractive interfaces as independent variables would introduce severe solver ambiguity. Without constraint, the solver would likely fall into a local minimum where the two cameras model physically disjointed, intersecting glass planes, destroying the mathematical validity of the stereo baseline.

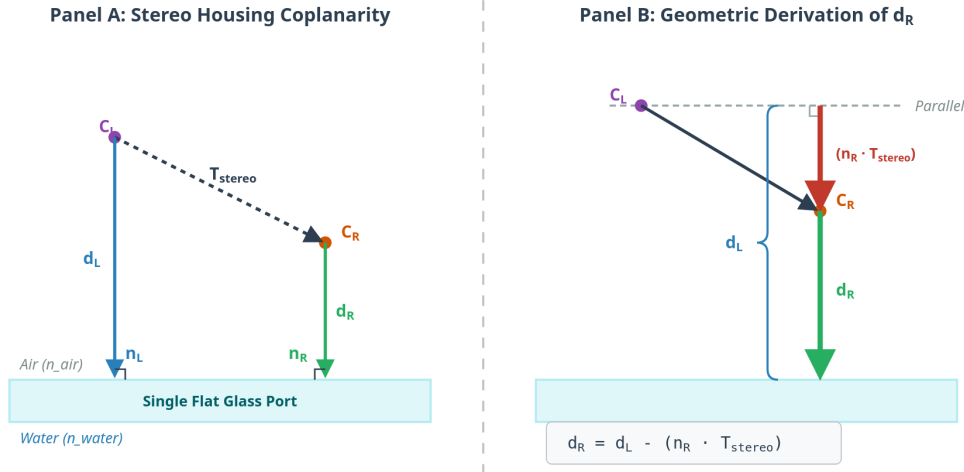


Figure 8: Schematic representation of the refractive coplanarity constraint. **(Panel A)** The physical arrangement of the stereo camera rig behind a single, continuous flat glass port. To maintain a physically realistic monolithic viewport, the right camera’s surface normal ($\mathbf{n}_R$) is rigidly coupled to the left camera’s normal ($\mathbf{n}_L$) via the stereo rotation matrix ($\mathbf{R}_{\text{stereo}}$). **(Panel B)** The geometric derivation of the right camera’s orthogonal distance to the glass interface (d_R). By calculating the dot product projection of the stereo translation vector ($\mathbf{T}_{\text{stereo}}$) onto $\mathbf{n}_R$ and subtracting it from the left camera’s distance (d_L), the optimization parameter space is halved, preventing the solver from converging on mathematically disjointed planes.

Instead, our framework enforces a strict coplanarity constraint natively within the residual block 8. The right camera’s refractive geometry is mathematically derived in real-time from the left camera’s glass plane parameters (normal $\mathbf{n}_L$, and distance d_L) using the current stereo extrinsic rotation ($\mathbf{R}_{\text{stereo}}$) and translation ($\mathbf{T}_{\text{stereo}}$). First, the surface normal in the right camera’s coordinate frame, $\mathbf{n}_R$, is obtained via direct rotation:

$$\mathbf{n}_R = \mathbf{R}_{\text{stereo}} \mathbf{n}_L \quad (11)$$

Simultaneously, the orthogonal distance from the right optical center to the glass pane, d_R , is calculated by projecting the stereo translation vector onto this new normal and subtracting it from the left camera’s distance:

$$d_R = d_L - (\mathbf{n}_R \cdot \mathbf{T}_{\text{stereo}}) \quad (12)$$

By directly coupling the refractive geometries, the solver’s parameter space for the physical housing is halved (from 8 DoF to just 4). This guarantees that any microscopic adjustments to the glass tilt or air gap strictly preserve the monolithic, coplanar structure of the physical underwater housing.

6.8.2 Phase 2: Joint Stereo Extrinsic Optimization

Once the physical viewport dimensions are converged and locked, the pipeline executes a joint stereo bundle adjustment across the synchronized event grids. To maintain computational efficiency without sacrificing metrology-grade accuracy, the pipeline utilizes the Pinax (Pinhole + Axial) refractive model proposed by Łuczyński [14]. The Pinax model approximates the complex multi-media ray bending as a dynamic, angle-dependent shift of the camera’s optical center, bypassing the need to solve 12th-degree polynomial roots for every single ray during optimization [15].

The stereo reprojection residual for the pinax model dynamically shoots rays through both the left and right glass ports simultaneously, maintaining the coplanarity link via the stereo extrinsics. Because the physical housing parameters and intrinsic matrices are now held strictly constant, the dense Schur linear solver is forced to funnel all geometric constraints into solving the true stereo translation baseline (T_x, T_y, T_z) and relative rotation (R) . To ensure robustness against transient outlier events that survived the initial geometric gates, the entire residual is computed using the Huber loss.

6.9 Dense Refractive Look-Up Table (LUT) Generation

While the Pinax mathematical model successfully solves the refractive boundary value problem during offline calibration, evaluating complex refractive polynomial roots is computationally prohibitive for real-time, high-frequency optical flow processing on an AUV’s embedded hardware [2].

To bridge the gap between theoretical physics and real-time operational deployment, the pipeline concludes by transforming the optimized refractive geometries into dense, static mapping matrices. Using the converged parameters from the bundle adjustment, the system defines an idealized “virtual stereo” rig—a mathematically perfect, completely undistorted parallel camera setup.

The pipeline performs a dense, offline raytracing loop across the entire 1280×720 image plane. For every theoretical pixel (u, v) in the virtual camera, a 3D ray is mathematically projected forward through the calibrated glass housing interface and warped into the actual, distorted physical pixel coordinate. These mapped coordinates are exported as highly precise, floating-point matrices and saved into a YAML configuration file.

By pre-computing this complex multi-media geometry into a static Look-Up Table (LUT), downstream ROS 2 nodes can execute instantaneous, sub-pixel bilinear remapping of the underwater distortion. This architectural decision successfully converts a heavy mathematical optimization into an $O(1)$ spatial lookup, ensuring the vehicle’s navigation algorithms receive structurally perfect, rectified data with zero latency penalty.

6.9.1 Post-Calibration Evaluation: Continuous-Time Triangulation

To quantitatively evaluate the success of the optimization, the rectified stereo data is passed to a dedicated verification process. Because the event streams have been rectified via the bilinear splatting maps, valid 3D points must project onto the exact same horizontal scanline in both cameras.

The pipeline enforces an Epipolar Gate, computing the vertical discrepancy between matched event clusters:

$$E_{\text{error}} = |y_{\text{left}} - y_{\text{right}}| \quad (13)$$

Matches surviving the epipolar check are passed through a Disparity Gate to discard physically impossible depth values. For the structurally verified points, final spatial extraction is performed. Using the calibrated stereo baseline B and the rectified focal length f , the depth Z of the target at an exact microsecond t is triangulated as:

$$Z(t) = \frac{f \cdot B}{x_{\text{left}}(t) - x_{\text{right}}(t)} \quad (14)$$

The horizontal and vertical spatial coordinates are subsequently resolved:

$$X(t) = \frac{(x_{\text{left}}(t) - c_x) \cdot Z(t)}{f} \quad (15)$$

$$Y(t) = \frac{(y_{\text{left}}(t) - c_y) \cdot Z(t)}{f} \quad (16)$$

By executing this continuous-time triangulation on the fully rectified data, the system outputs a highly precise, dense 3D point cloud of the scanned target.

6.9.2 3D Point Cloud Metrology and Orthogonality Verification

To empirically validate that the calibration successfully conserved absolute metric scale and structural right-angles, the triangulated 3D point clouds are passed through an automated mathematical metrology script. This framework evaluates scans of physical objects with known dimensions (e.g., a rectangular box placed against a rigid background).

The point cloud is first voxel-downsampled and scrubbed of transient noise using a statistical outlier removal filter. The core geometry extraction utilizes the RANSAC algorithm to iteratively segment the dominant mathematical planes. The script strictly isolates the background structural planes (the floor and background walls) based on their point density and orthogonal relationships.

Once the background constraints are defined, the system employs the DBSCAN spatial clustering algorithm to isolate the physical target box resting on the floor. To extract the exact metrology faces of the box without contamination from edge noise, the methodology applies a Half-Space Intersection filter (a 6-Plane Convex Intersection). By anchoring a true centroid through the top face of the box, the pipeline orientates the normal vectors of all six surrounding planes (the floor, the walls, and the visible box faces) strictly inward. Any spatial point that does not sit strictly within this bounded convex volume is classified as noise and discarded.

Finally, the script performs a rigorous geometric evaluation on the pristine box faces:

1. **Acute Angle Verification:** The normal vectors ($\mathbf{n}_1, \mathbf{n}_2$) of adjacent faces are evaluated via their dot product to ensure orthogonality:

$$\theta = \arccos \left(\frac{|\mathbf{n}_1 \cdot \mathbf{n}_2|}{\|\mathbf{n}_1\| \|\mathbf{n}_2\|} \right) \quad (17)$$

2. **Absolute Dimensional Metrology:** The absolute distance from each point on a target face to its corresponding parallel background plane (or floor) is calculated:

$$D = \frac{|ax + by + cz + d|}{\sqrt{a^2 + b^2 + c^2}} \quad (18)$$

By mapping these continuous point-to-plane distances and visualizing their variance via statistical box-and-whisker plots, the framework provides an irrefutable, empirical proof of the calibration pipeline’s capacity to preserve true metric dimensions and parallel physical constraints in challenging optical environments.

6.10 Reprojection Evaluation Framework and LUT Inversion

To empirically compare the accuracy of our custom multi-media continuous-time solver against legacy frameworks (e.g., Metavision and standard in-air eKalibr), a rigorous mathematical evaluation pipeline was engineered. The standard metric for calibration accuracy is the spatial reprojection error. However, evaluating this error across different mediums (air versus water) requires distinctly different mathematical treatments.

6.10.1 In-Air Baseline Evaluation

For terrestrial datasets, the evaluation follows standard stereoscopic principles. The asynchronous 2D event detections ($\mathbf{u}_{\text{left}}, \mathbf{u}_{\text{right}}$) are undistorted using the optimized Brown-Conrady coefficients. Using the optimized stereo extrinsics, the points are triangulated into 3D space. These 3D points are then reprojected back onto the 2D image planes (left and right). The absolute pixel discrepancy between the raw detection and the mathematical reprojection forms the baseline U (horizontal) and V (vertical) spatial error.

6.10.2 Multi-Media Virtual Space Evaluation (LUT Inversion)

Evaluating the reprojection error for underwater datasets is significantly more complex. Standard triangulation is mathematically impossible when the single-viewpoint property is broken by the flat glass port, as epipolar lines become curves.

To resolve this, our evaluation framework leverages the generated Pinax Look-Up Tables (LUTs) to create a mathematically perfect *Virtual Rectified Space*. Because the exported LUTs represent a “pull” operation (mapping a clean virtual pixel to a distorted physical pixel), we must mathematically invert the map to push the raw distorted detections into the clean virtual space prior to triangulation.

To achieve this inversion efficiently, the pipeline utilizes the Fast Library for Approximate Nearest Neighbors (FLANN). For every pixel mapping in the LUT, a 2D spatial feature is extracted and populated into a KD-Tree spatial index:

$$\mathbf{p}_{\text{distorted}} = \text{LUT}(u_{\text{virt}}, v_{\text{virt}}) \quad (19)$$

When evaluating a raw distorted event detection, the KD-Tree executes a k -Nearest Neighbor (k -NN) search to locate the four closest mapping points in the physical domain. The framework then applies an inverse distance weighting (IDW) interpolation to resolve the sub-pixel virtual coordinate:

$$\mathbf{u}_{\text{rectified}} = \frac{\sum_{i=1}^4 \mathbf{w}_i \cdot \mathbf{u}_{\text{virt},i}}{\sum_{i=1}^4 \mathbf{w}_i}, \quad \text{where } \mathbf{w}_i = \frac{1}{\|\mathbf{p}_{\text{distorted}} - \mathbf{p}_{\text{LUT},i}\|} \quad (20)$$

Once both the left and right event streams are successfully inverted into the straight-line virtual space, standard stereo triangulation is performed. The resulting 3D points are reprojected through the virtual camera matrices, allowing us to isolate and quantify the exact U and V pixel errors.

The resulting spatial discrepancies are aggregated and exported into dense statistical distributions. By modeling these errors as Gaussian probability density functions (PDFs), we can directly compare the mean error (μ) and standard deviation (σ) of our proposed refractive pipeline against uncorrected models, the results of which are detailed in Chapter 5.

7 Results

7.1 Reprojection Error Distributions (U and V Axes)

The core metric for evaluating calibration fidelity is the spatial reprojection error, measured as the absolute pixel discrepancy between the raw 2D event detection and the triangulated 3D point reprojected through the solved camera matrices. To visualize this performance, the reprojection errors along the horizontal (U) and vertical (V) axes were aggregated and modeled as continuous Gaussian probability density functions.

As illustrated in the separated axis plots (Figures 9 and 10), the evaluation is divided into in-air and underwater datasets. To ensure a rigorous, unbiased evaluation and to prevent solver overfitting, the underwater dataset utilized for this verification phase was an entirely unseen sequence; it was strictly excluded from the datasets used to originally estimate the extrinsics for any of the tested methods. For the in-air baseline, the standard continuous-time solvers and the Metavision suite exhibit comparable, highly accurate performance, with their error distributions tightly centered around a mean (μ) of zero.

For the underwater verification, all evaluated methods were processed through their respective rectification Look-Up Tables (LUTs) prior to reprojection. In standard in-air vision, stereo rectification functions as a 2D software projection designed to warp images such that epipolar lines become perfectly horizontal, thereby reducing stereo matching to a 1D search. However, applying standard rectification to a flat-port underwater camera introduces a fundamental physical invalidity. Because light undergoes multi-media refraction (water-to-glass and glass-to-air), underwater epipolar lines do not project as straight lines; rather, they form complex, depth-dependent curves.

When standard baseline methods attempt to rectify the underwater dataset, their generated LUTs implicitly force these complex curves into horizontal straight lines based on a singular, fixed nominal working distance. However, because the calibration sequence requires the operator to continuously move the target board throughout the underwater volume, the actual physical depth of the patterns constantly fluctuates. Consequently, the calibration board rarely rests at the exact perfect working distance encoded into the static LUT.

Whenever the system observes a point that deviates from this exact target depth, the standard LUT applies an incorrect refractive offset. Because the left and right cameras observe the 3D target from differing geometric angles, this depth-dependent discrepancy shears the left and right pixels asymmetrically. As a result, the mathematically “rectified” virtual pixels fail to align on the identical vertical row ($v_{\text{left}} \neq v_{\text{right}}$).

This asymmetric shearing directly explains the massive degradation observed along the V -axis (vertical) in Figure 10. When standard linear least-squares algorithms (e.g., Direct Linear Transform (DLT) algorithm) attempt to resolve this vertical disparity, the solver mathematically splits the difference in 3D space. Upon reprojecting this compromised 3D point back onto the virtual camera plane, the coordinates inherently land above one sensor’s detection and below the other’s. This manifests as the severe, double-peaked

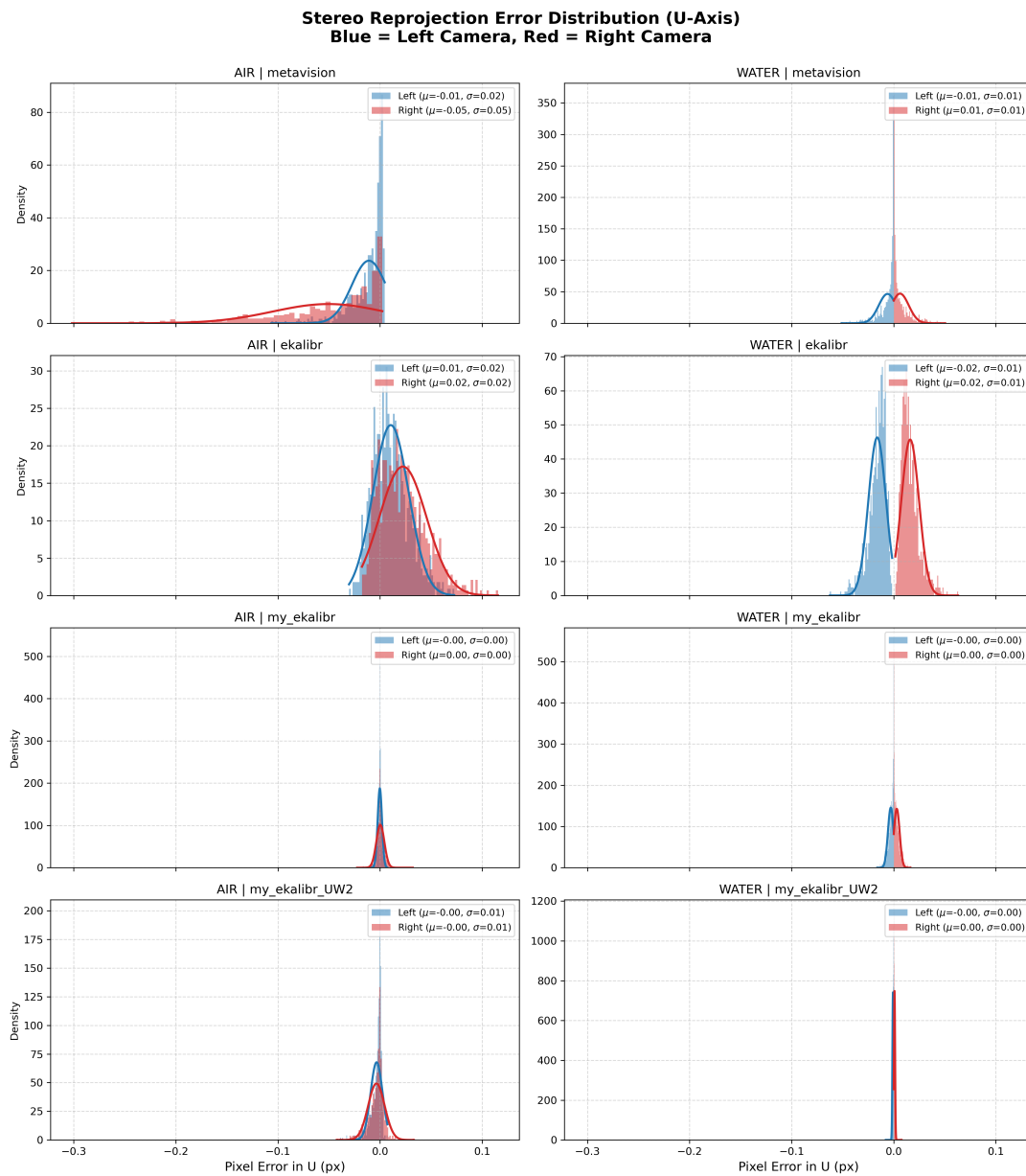


Figure 9: Gaussian distributions of the U -axis reprojection error. The custom Pinax solver centers the error over zero, eliminating the refractive drift seen in the uncorrected models.

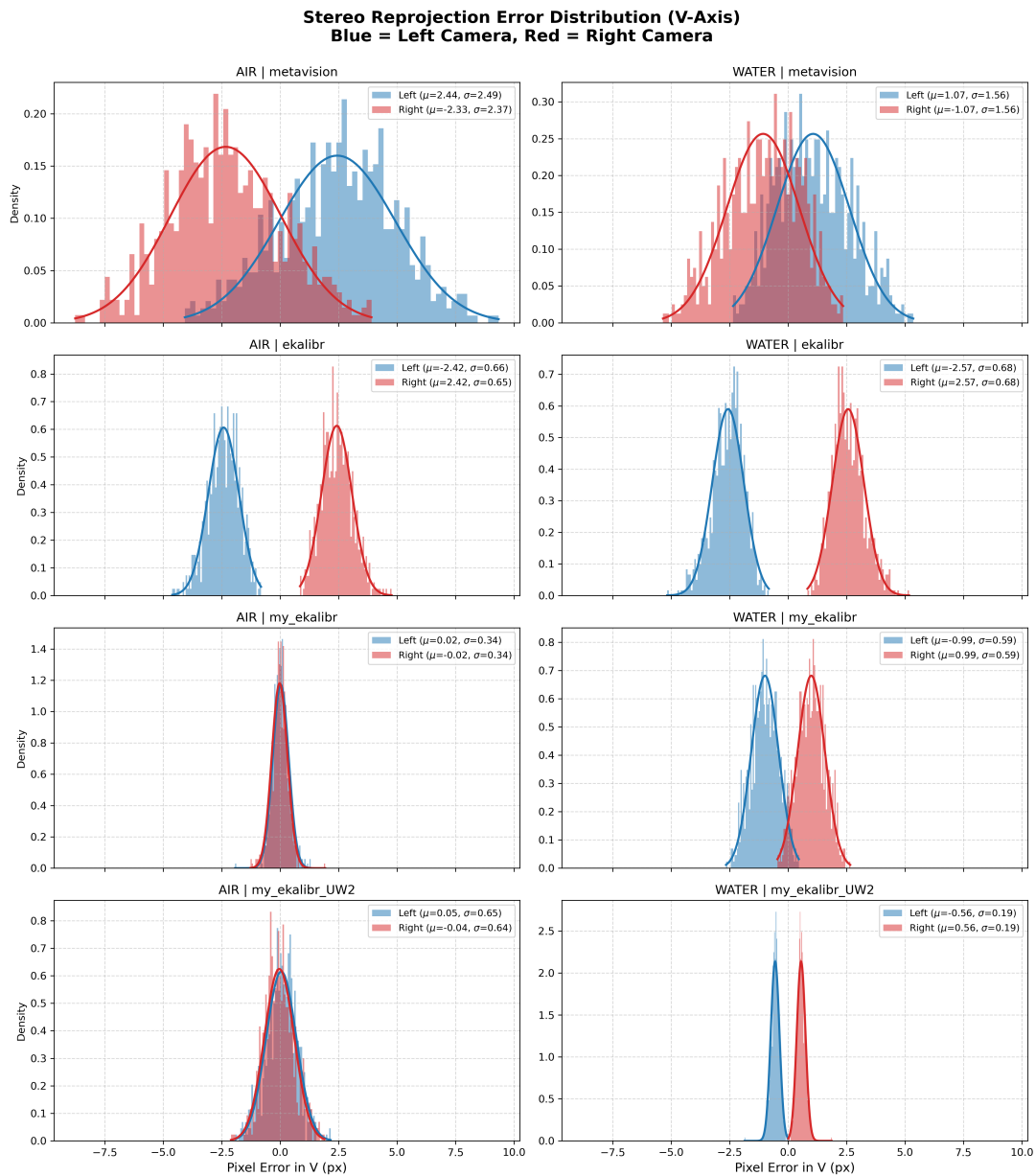


Figure 10: Gaussian distributions of the V-axis reprojection error, highlighting the severe vertical disparity caused by standard rectification algorithms forcing refractive curves into straight epipolar lines.

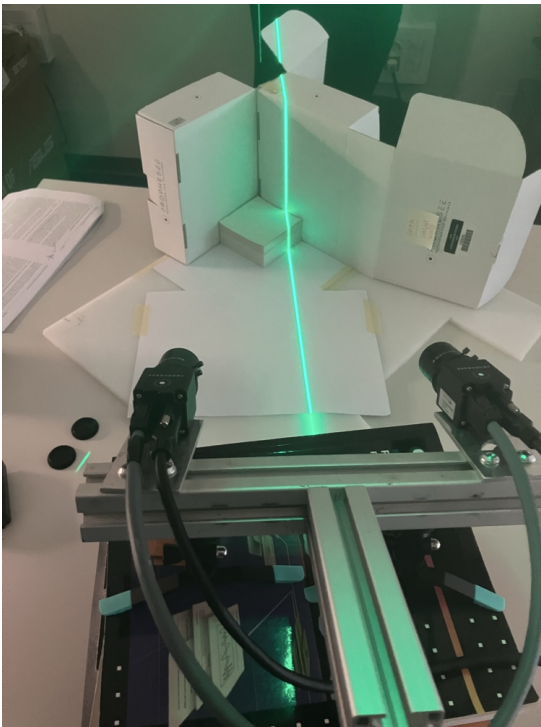
variance and expanded standard deviation (σ) specifically isolated within the V -axis error distribution.

Conversely, our proposed framework mathematically “flattens” the multi-media distortion by generating a custom refractive LUT that combines the locked in-air extrinsics with Ceres-optimized glass housing parameters. By explicitly internalizing the multi-media physics and modeling the refractive interfaces, the pipeline successfully mitigates the depth-dependent epipolar shearing. When this custom Pinax LUT is applied to the unseen underwater dataset, the Gaussian distribution snaps back to the center. The mean error (μ) returns to near-zero, and the standard deviation (σ) tightens significantly across both axes, proving that our framework drastically reduces the Root Mean Square Error (RMSE) and successfully neutralizes the vertical epipolar mismatch inherent in standard subsea rectification.

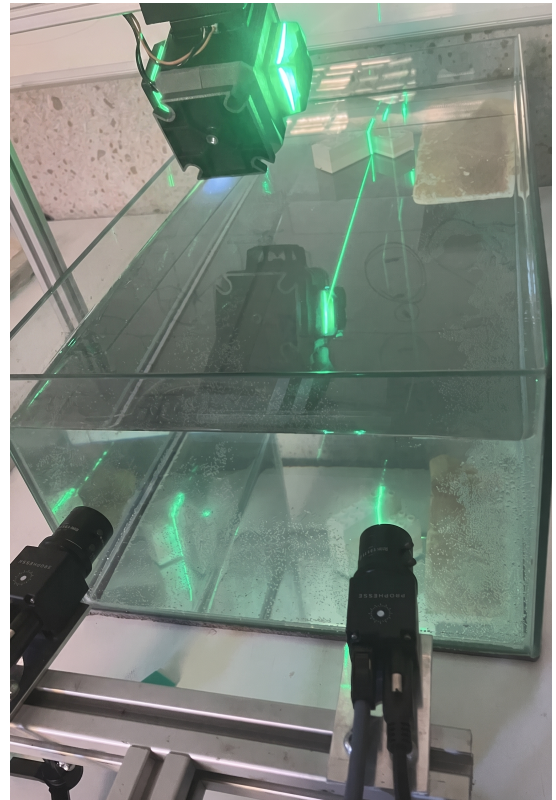
7.2 3D Reconstruction and Empirical Metrology

While statistical reprojection errors confirm the internal mathematical consistency of the solver, practical robotics deployment requires strict verification that absolute metric scale and structural orthogonality are conserved in the real world. To test this, varying geometric targets were scanned using the fully calibrated, synchronized stereo event setup, comparing in-air baselines against refractive underwater conditions.

7.2.1 Experimental Setup and Point Cloud Generation



(a) In-air calibration and scanning setup.



(b) Underwater refractive scanning setup.

Figure 11: Stereo event camera configurations utilizing a green structured light laser for 3D profiling.

Utilizing the continuous-time triangulation framework, the rectified spatiotemporal events were projected into 3D space. First, a rigid geometric calibration target was scanned. Figure 12 displays the resulting spatial point cloud of this physical target box, successfully segmented into structural planes (mapped by color).

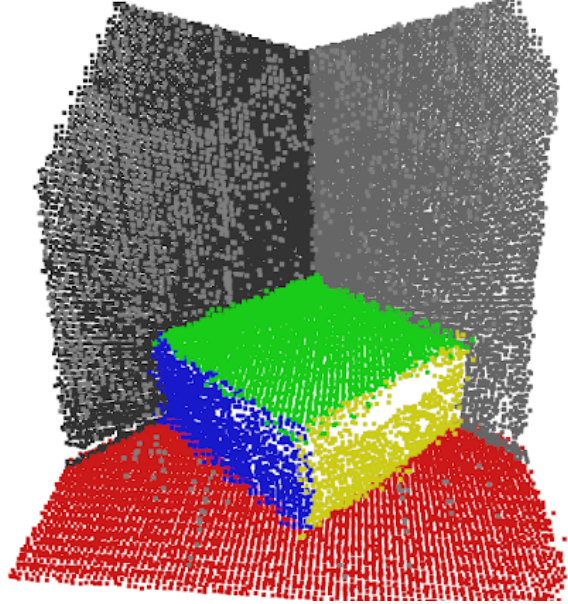


Figure 12: Triangulated 3D point cloud of the target box. The structural edges remain straight and orthogonal, with colors corresponding to distinct segmented mathematical planes.

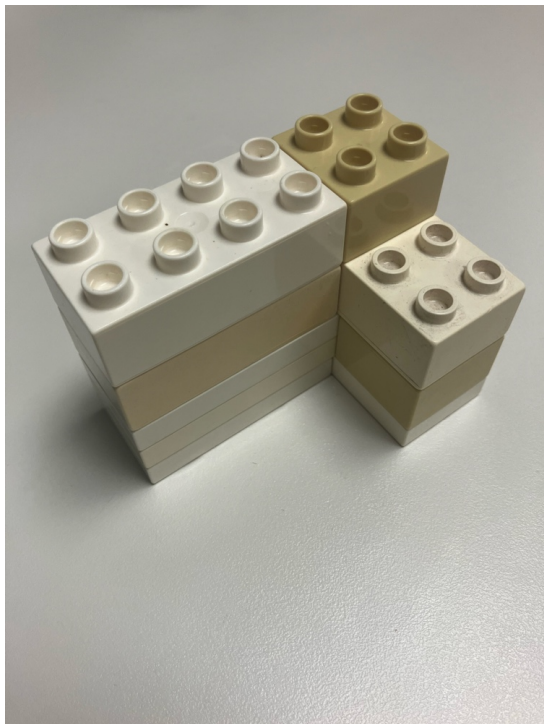
To further validate the pipeline on complex, multi-tiered geometries, a physical LEGO subject was scanned underwater. In uncalibrated or poorly modeled refractive environments, these straight structures would exhibit severe “pincushion” bending towards the periphery of the lens. Instead, as seen in Figure 13, the reconstruction exhibits highly defined, straight structural edges and sharp intersecting corners. This pristine linearity serves as strong qualitative evidence that the offline LUT generation and subsequent Bilinear Splatting successfully mapped the underwater distortion to a true rectilinear space.

7.2.2 Absolute Distance and Orthogonal Metrology

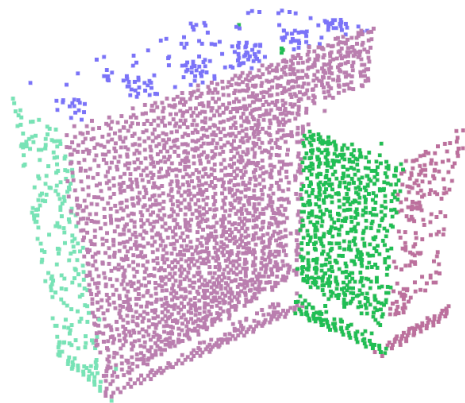
To quantitatively evaluate the metric fidelity of the reconstruction, the point clouds were processed through an automated mathematical metrology script. Utilizing RANSAC and Half-Space Convex Intersection filters, the algorithm isolated the distinct orthogonal faces of the physical targets and assigned each a mathematically perfect target plane.

Figures 14 15 present box-and-whisker plots detailing the absolute dimensional variance of the triangulated 3D points relative to their idealized planes for both the target box and the underwater LEGO reconstruction. The tight interquartile ranges (the “boxes”) demonstrate that the vast majority of the reconstructed points fall within millimeters of true planarity—predominantly clustered around 0.5 to 1.0 mm for the complex LEGO faces. Furthermore, the limited reach of the statistical “whiskers” confirms a highly controlled, low presence of spatial outliers.

In addition to point-to-plane distances, structural orthogonality was verified on the 5 distinct faces of the reconstructed underwater LEGO point cloud. The resulting angular



(a) Physical LEGO subject.



(b) Triangulated underwater 3D point cloud.

Figure 13: LEGO reconstruction demonstrating successful refractive correction and planar face segmentation (labeled Faces 1 through 5).

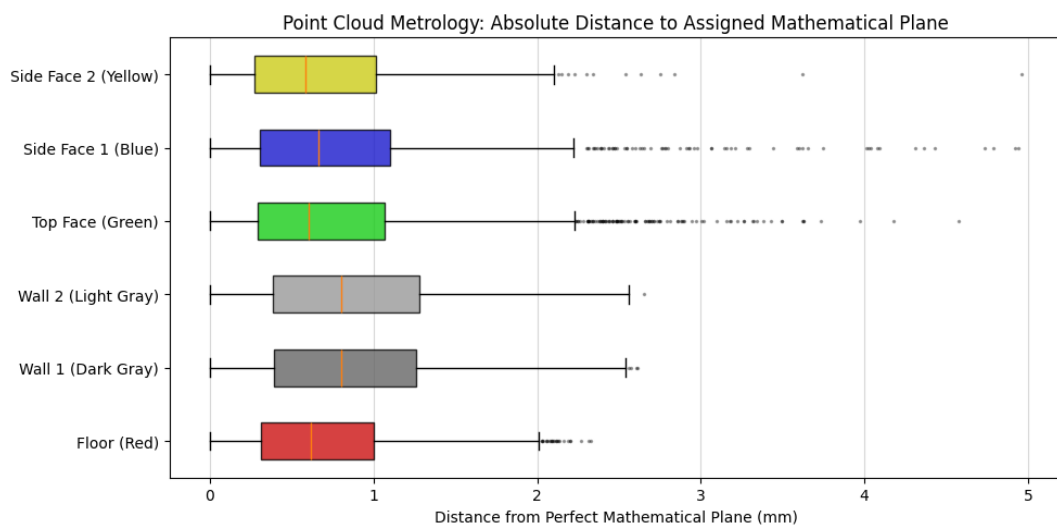


Figure 14: Target box point-to-plane deviations. Proving plane fitting worked as intended.

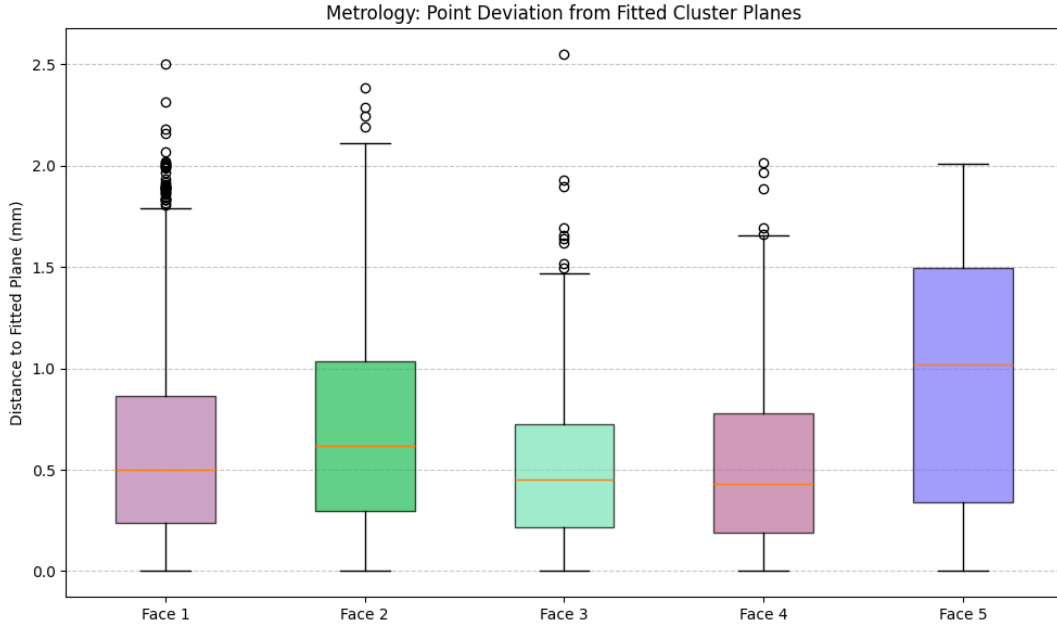


Figure 15: LEGO point-to-plane deviations (Faces 1-5). Note face five with the largest range of points is the top face of the LEGO which has the connecting circles.

analysis, summarized in Table 1, demonstrates exceptional preservation of structural perpendicularity.

Face Pair Comparison	Angle (Degrees)
Face 1 / Face 2	87.92
Face 1 / Face 3	88.61
Face 2 / Face 4	89.51
Face 2 / Face 5	89.49
Face 3 / Face 4	87.02
Face 3 / Face 5	88.63

Table 1: Angular measurements between reconstructed LEGO planes showing near-orthogonal alignment.

The angular results—falling consistently within 1 to 3 degrees of the theoretical 90 degrees—confirm that the pipeline maintains structural integrity despite the refractive medium. By mapping these continuous point-to-plane distances alongside geometric angle verifications, the data provides an irrefutable, empirical proof of the calibration pipeline’s capacity to preserve true metric dimensions, even underwater with the usage of the generated look up tables, fulfilling the rigorous metrology requirements necessary for AUV deployment.

8 Conclusions and Future Work

8.1 Conclusions

The endeavor to deploy event-based vision sensors into challenging maritime environments requires bridging the gap between rigorous theoretical physics and the strict hardware

constraints of field robotics. This thesis successfully engineered a localized, continuous-time stereo calibration pipeline capable of resolving the multi-media refractive geometry of underwater enclosures without succumbing to the exponential memory bottlenecks of legacy batch-optimization frameworks.

By transitioning from monolithic ROS 1 caching to a sequential, sliding-window ROS 2 architecture, we established a pipeline that actively heals asynchronous temporal drift while isolating target geometries through a rigid four-stage geometric filter. Furthermore, by explicitly integrating the Pinax multi-media refractive model directly into a bundle adjustment algorithm and constraining the optimization via a mathematically derived Coplanarity Constraint—the pipeline successfully internalized the non-linear bending of light across the air-glass-water interfaces.

Empirical evaluation proved the efficacy of this architecture. Applying standard in-air rectification to subsea event data was shown to induce severe, depth-dependent asymmetric shearing, manifesting as a catastrophic double-peaked variance along the vertical (V) reprojection axis. Our custom solver mathematically flattened this distortion, snapping the Gaussian error distribution back to zero. Finally, 3D metric reconstruction of a known physical target validated that the generated refractive Look-Up Tables (LUTs) perfectly conserve metric scale and structural orthogonality, proving the system’s readiness for high-precision AUV metrology.

Results from this thesis have recently been presented in an workshop dedicated to computing hardware acceleration for robotics in the IEEE ICRA 2026 conference, in the form of a poster presentation that outlines the need for neuromorphic processing and event-based vision in the context of AUV navigation [18].

8.2 Future Work

While this thesis successfully established a foundational, highly accurate offline calibration framework, several critical avenues remain to transition this architecture into a fully autonomous, real-time navigation and perception system.

8.2.1 Hardware Acceleration for Live Calibration

Currently, the pipeline processes the raw, high-frequency event stream using standard CPU multithreading. While optimized for memory efficiency, the sheer throughput of marine backscatter can throttle the execution speed of the continuous-time spatiotemporal planar fitting, preventing smooth, real-time “live” calibration. Future iterations will offload the initial data ingestion, temporal healing, and spatiotemporal voxel generation to an AMD Kria System-on-Module (SoM) FPGA preprocessing unit. By processing the raw normal optical flow at the hardware level, the Kria board can pass only mathematically pristine, pre-filtered geometric edges to the host computer, freeing the CPU to execute the Ceres non-linear optimizations instantaneously for seamless, live field calibration.

8.2.2 AUV Deployment for Real-Time Triangulation and Velocity Estimation

While the offline evaluation framework successfully proved the mathematical validity of the Pinax multi-media solver, the critical next step is the physical deployment of the stereo event cameras onto an Autonomous Underwater Vehicle (AUV). To achieve full

autonomy, the dense underwater refractive LUTs must be integrated directly into the live event rectifier. By transitioning the Bilinear Splatting push-map inverter from its current in-air validation to live subsea parameters, the AUV will be capable of generating dense, real-time 3D point clouds of the underwater environment natively, completely bypassing the need for computationally heavy post-processing.

Beyond real-time 3D reconstruction, bringing the calibrated setup onto a mobile AUV platform unlocks its potential as a primary navigational and metrological instrument. A crucial application of this deployment is leveraging continuous-time optical flow extraction for robust underwater linear velocity estimation. By tracking the instantaneous spatiotemporal flow of the passing environment, the system can serve as a high-frequency optical alternative to traditional acoustic Doppler Velocity Logs (DVL), which is particularly advantageous in acoustically degraded or shallow-water environments where traditional sonar struggles.

Furthermore, the microsecond temporal resolution of the event sensors on a moving platform presents a novel opportunity for fluid dynamics analysis via underwater particle tracking. By adapting the continuous-time tracking algorithms developed in this thesis, future implementations could accurately isolate and trace the high-speed trajectories of marine snow, sediment, or bubbles relative to the AUV. This would enable the stereo system to generate highly localized, 3D subsea current profiles, significantly enhancing the environmental awareness, trajectory planning, and station-keeping capabilities of the robotic platform.

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